

November 12th, 2025
Editors of *Nature Biotechnology*

Dear Editors,

We are pleased to submit our manuscript entitled “*Leveraging Large Language Models for Redundancy-Aware Pathway Analysis and Deep Biological Interpretation*” for consideration by *Nature Biotechnology*. This work presents MAPA, a novel computational framework that integrates Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to address a long-standing and pervasive problem in functional interpretation of omics data: pathway redundancy and fragmented annotation in enrichment analysis.

Pathway analysis is widely used in systems biology to uncover biological mechanisms from vast and complex omics data. However, conventional pathway analysis tools often return long, highly overlapping lists of pathways that are difficult to interpret and integrate into a cohesive biological narrative. MAPA systematically resolves this bottleneck through a two-stage process:

(1) Redundancy-aware functional module identification:

MAPA first embeds enriched pathways into a semantic space using text-transformer models. It then constructs a similarity network and applies clustering methods to group functionally similar pathways into coherent functional modules. This step transforms dozens or even hundreds of redundant pathways into unified biological functional units.

(2) Functional module annotation based on LLM and RAG:

Each functional module is then annotated using an RAG framework to reduce hallucinations. MAPA dynamically retrieves the most relevant biological descriptions of member pathways and prompts an LLM to generate accurate and concise module annotations grounded in domain knowledge. This approach not only enhances interpretability but also ensures reproducibility and transparency, as all intermediate retrievals and generations are recorded.

Our benchmarking studies show that MAPA outperforms existing tools in both functional module identification (Adjusted Rand Index = 0.95) and annotation quality (highest agreement with human expert labels across eight curated modules). To further demonstrate real-world utility, we applied MAPA to a comprehensive multi-organ monkey aging dataset, spanning transcriptomics across 78 tissues, plasma proteomics, and plasma metabolomics. MAPA uncovered robust and interpretable patterns of age-related dysregulation, such as ferroptosis vulnerability along the brain-gut axis and systemic downregulation of FOXO signaling, not captured by traditional enrichment approaches.

As LLMs continue to transform biomedical research, a growing body of work has explored their use for gene set annotation (Hu *et al.*, *Nature Methods*, 2024; Wang *et al.*, *Nature Methods*, 2025), disease modeling (Li *et al.*, *Nature Medicine*, 2025), and so on. Here, based on LLMs and RAG, MAPA offers a principled, task-specific, and interpretable solution to a core methodological challenge in omics: the functional redundancy of pathway analysis results.

Given *Nature Biotechnology*’s focus on advancing biotechnology across the biological,

biomedical, and computational sciences, we believe this work will resonate with a broad readership, particularly in computational biology, systems biology, and omics data analysis. Our approach offers innovative computational methods that enhance the interpretation of multi-omics data, a topic of relevance to the fields of large-scale biology, bioinformatics, and the broader biotechnology community. This manuscript has not been published or submitted elsewhere, and all authors have read and approved the final version. A complete set of code and documentation is available via GitHub (<https://github.com/jaspershen-lab/mapa>) and MAPA website (<https://www.shen-lab.org/mapa-website/>), ensuring full reproducibility and transparency. A web-based MAPA application is also available at: <https://mapashiny.jaspershenlab.com/>. To facilitate the review process, we are happy to provide temporary OpenAI and SiliconFlow API keys for reviewers to run MAPA:

OpenAI API key (for local version of MAPA and mapashiny):

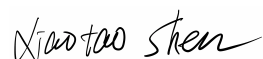
sk-proj-mf0lfCIHY4pUFJ94GPydG77ghJoaOd-oFUrflmW_bJpt0fMuw-
pk43Lz_Dv2TsJqZ16cThx92FT3BlbkFJT9dzQepYrw0bDdduXqKqg60On5wCEStgiCk4jnn
a5B1Fz1Mpq-N9IutkMA1sRVdMJwZoW2TkMA

SiliconFlow API key (for online version mapashiny):

sk-ybmjbxlvupgvmbaqievpkntlinzeptnxmubsqsetgincxgbw

Thank you for your consideration and look forward to your response.

Yours sincerely,



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Potential Reviewers

Include:

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